

SOFTWARE ENGINEERING LAB – 4

VIBE CODING PROMPTS

Project: Fruit Ninja – Python/Pygame

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Task 1 – Refine Collision Detection

Improve the collision detection in the Fruit Ninja game so that fruit slicing works reliably, especially when the player performs fast mouse swipes. Use the line segment between the previous mouse position and the current mouse position for collision detection instead of checking only the current mouse position. Also add a small collision margin around the fruit so that fast swipes are detected reliably. Do not change unrelated gameplay features.

Task 2 – Implement Game Over Screen

Implement a proper Game Over screen for the Fruit Ninja game. When the player loses, stop the normal gameplay and display a clear Game Over message/state to the player. The Game Over state should integrate cleanly with the existing game flow and should not affect unrelated functionality.

Task 3 – Add Replay and Difficulty Selection

Add replay functionality and difficulty selection to the Fruit Ninja game. After Game Over, provide options for Easy, Medium, Hard, and Exit. Make the difficulty levels meaningfully different by changing gameplay parameters such as fruit spawn interval, bomb probability, and object speed. Allow the player to start a new game without restarting the application. Keep the existing gameplay and collision improvements working correctly.

Task 4 – Add Sound Effects

Add sound effects to the Fruit Ninja game for the major gameplay events. Implement a sound management module and add separate audio feedback for fruit slicing, bomb explosion, and Game Over. Ensure the sounds are triggered at the correct events and do not break the existing gameplay, replay, difficulty selection, or collision detection features.